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RESEARCH ARTICLE

HEDP-Modified Cu_2O Catalyst Enables Efficient C_{2+} Product Formation in Acidic CO_2 Electroreduction

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Received: 12 August 2025 | **Revised:** 19 February 2026 | **Accepted:** 9 March 2026

Keywords: acidic CO_2 electroreduction | C_{2+} products | hydroxyethanediphosphonic acid (HEDP) | surface modification

ABSTRACT

Efficient electrocatalytic CO_2 reduction (ECR) to multi-carbon (C_{2+}) products under acidic conditions requires suppression of the competing hydrogen evolution reaction, which can be achieved with high concentrations of potassium ions (K^+) dissolved in the electrolyte. However, this excess of salt ions generates unwanted byproducts, negating the advantage of efficiency over extended use. Herein, we demonstrate a strategy to modify Cu_2O with hydroxyethanediphosphonic acid (HEDP) to induce a high K^+ microenvironment, thereby enabling efficient CO_2 conversion to C_{2+} products, even at low K^+ concentrations. In a 3.0 M K^+ electrolyte (pH 1.0), a faradaic efficiency (FE) of 79.3% for C_{2+} products was achieved at a current density of −700 mA cm^{-2}. Notably, after activation in a K^+-containing electrolyte, the FE for C_{2+} products can still reach 50% even in a K^+-free electrolyte (1 mM H_2SO_4), while the FE for H_2 remains below 15%. Experimental and first-principles calculations revealed that HEDP facilitated K^+ enrichment on the catalyst interface, induced a higher local pH microenvironment, and promoted C−C coupling. Our interfacial microenvironment optimization strategy harnesses the critical role of the electrode-electrolyte interface in promoting efficient CO_2 electroreduction to C_{2+} products under acidic conditions with low K^+ concentrations.

1 | Introduction

Electrocatalytic CO_2 reduction (ECR), which utilizes renewable electricity to convert CO_2 into high-value-added chemicals, offers a sustainable and carbon-net-zero solution to global energy and climate challenges [1–7]. Currently, ECR of multi-carbon (C_{2+}) products is generally performed in neutral or alkaline aqueous electrolytes to suppress the competing hydrogen evolution reaction (HER) and enhance C−C coupling on the catalyst surface [8–10]. However, in alkaline and neutral electrolytes, the strong local alkalinity induced by ECR causes CO_2 to combine with hydroxide ions (OH^-) to form (bi-) carbonates. This exergonic and spontaneous process releases approximately −56 kJ mol^{-1} of energy [11] and limits the single-pass conversion efficiency (SPCE) of CO_2 to C_{2+} products to below 25% [12, 13]. Performing ECR in acidic electrolytes has been proposed to address these issues.

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Under acidic conditions, the high concentration of protons in the electrolyte suppresses the formation of (bi-) carbonate and prevents CO_2 crossover. Nevertheless, owing to the high availability of protons and fast kinetics of the competitive HER, catalysts in acidic electrolytes tend to favor HER, thereby limiting the formation of C_{2+} products [14, 15]. Pioneering studies have shown that the addition of alkaline cations to acidic electrolytes effectively addresses this problem [16]. Since then, acidic electrolytes with high concentrations of K^+ have been widely reported to promote C−C coupling and inhibit HER. For example, Sargent et al. [17] constructed hydrophilic-hydrophobic nanochannels carrying cations using covalent organic framework particles (COF) and cation-exchange perfluorosulfonic acid (PFSA). The Cu catalyst co-modified with COF and PFSA achieved a faradaic efficiency (FE) of 75% for C_{2+} products in a 3.0 M KCl (pH ≈ 1.0) electrolyte. Huang et al. [18] found that the FE for C_{2+} products produced by electrochemically reduced Cu porous nanosheets can reach 83.7% in a 3.0 M KCl + 0.05 M H_2SO_4 electrolyte. Although previous studies have shown that the presence of high concentrations of alkaline cations in the electrolyte helps to buffer the local pH and promotes the generation of C_{2+} products, it also presents certain challenges. The high local pH, along with the accumulation of K^+ on the catalyst surface, still led to the formation of salt precipitates that blocked the CO_2 mass transfer channels, increased the interfacial resistance, and weakened the stability of the catalytic system [19]. Therefore, the effect of K^+ concentration reflects a balance between interfacial modulation and system stability. This balance also informs the design of acidic ECR systems toward the efficient and stable production of C_{2+} products.

Conventional strategies to enrich K^+ in the electric double layer employ modifiers such as sulfonate groups or polyionic liquids, each of which typically binds only a single cation. In comparison, phosphonate groups can engage in multidentate coordination with alkali cations, offering enhanced potential for local K^+ association compared with typical sulfonate-based modifiers. Previous studies have reported that hydroxyethanediphosphonic acid (HEDP) forms weak complexes with K^+ (log $K \approx 1.8$) [20], indicating a thermodynamically detectable yet not particularly strong interaction. Such moderate coordination may affect the interfacial ionic environment. However, the reported stability constant reflects equilibrium thermodynamics and does not directly provide insight into the kinetic behavior under electrochemical conditions. On this basis, we designed and synthesized HEDP-modified Cu_2O ($Cu_2O@HEDP_{0.8}$), in which the HEDP (Figure S1) is strongly anchored to Cu sites. Our $Cu_2O@HEDP_{0.8}$ catalyst achieved an FE of 79.3% for C_{2+} products in an electrolyte with a K^+ concentration of 3.0 M (pH 1.0), which was 1.7 times higher than that of bare Cu_2O (b-Cu_2O). At a current density of −700 mA cm^{-2}, the FE of H_2 over $Cu_2O@HEDP_{0.8}$ was only 5.6%, which is lower than the 13% over b-Cu_2O, indicating that the HEDP modification effectively suppressed the HER. Furthermore, after activation in a K^+-containing electrolyte, even in a K^+-free electrolyte, the FE for C_{2+} products over $Cu_2O@HEDP_{0.8}$ could still reach as high as 50%, whereas the FE for H_2 remained below 15%. Mechanistic investigations revealed that the phosphate groups in the HEDP molecules effectively enhance the local K^+ on the catalyst intersurface. This work introduces phosphonate groups as additional sites for regulating the microenvironment, thereby inducing higher local pH values to promote *CO protonation and subsequent C−C coupling. This enables efficient formation of the C_{2+} product even in low K^+ electrolytes.

2 | Results and Discussion

2.1 | Material Synthesis and Characterizations

$Cu_2O@HEDP_{0.8}$ and b-Cu_2O were synthesized by chemical reduction at ambient temperature and pressure. The morphologies of the catalysts were characterized using scanning electron microscopy (SEM) and high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM). SEM images showed that both $Cu_2O@HEDP_{0.8}$ and b-Cu_2O displayed nanoparticle morphology (Figure S2), indicating that the HEDP modification did not significantly affect the morphology of Cu_2O. TEM and HAADF-STEM further revealed that the $Cu_2O@HEDP_{0.8}$ catalyst contained irregular nanoparticles (Figure 1a) with an average particle size of 31 ± 7.8 nm determined from statistical analysis of 100 individual particles (Figure S3). Fast Fourier transform (FFT) analysis of the region within the blue box confirms the high crystallinity of Cu_2O (Figure 1b). The lattice fringes of 0.17 and 0.30 nm corresponds to the (211) and (100) lattice spacings of Cu_2O, respectively [21]. Through energy-dispersive X-ray spectroscopy (EDS) element mapping and energy spectrum analysis, it was observed that P was present in $Cu_2O@HEDP_{0.8}$ and was uniformly distributed (Figure 1c). This further confirms the uniform distribution of HEDP on the Cu_2O surface.

Fourier-transform infrared (FT-IR) spectroscopy was conducted to investigate the surface chemical properties of $Cu_2O@HEDP_{0.8}$ and b-Cu_2O. As shown in Figure 1d, the bands at 1107 and 1055 cm^{-1} can be attributed to the symmetric stretching vibration of the P═O bond and the antisymmetric stretching vibration of the P−O bond, respectively [22, 23], whereas the weak band near 2855 cm^{-1} is associated with the C−H stretching vibration of the CH_3 group in the HEDP molecule [24]. Compared to b-Cu_2O, the $Cu_2O@HEDP_{0.8}$ sample exhibited characteristic bands for ν(C−H), ν(P═O), and ν(P−O), indicating that HEDP was introduced onto the Cu_2O surface. X-ray photoelectron spectroscopy (XPS) was conducted to analyze the electronic states of the samples (Figure 1e; Figure S4). In contrast to b-Cu_2O, the characteristic peak of P 2p at ∼133 eV indicates the presence of P element in $Cu_2O@HEDP_{0.8}$ (Figure S4a). Further inspection of the high-resolution P 2p spectrum revealed peaks at 132.8, 133.6, and 135.1 eV corresponding to P═O, P−C, and P−O bonds, respectively (Figure 1e) [25, 26]. The presence of chemical bonds between P and O was also confirmed by the O 1s spectrum. Specifically, the P−O bond of the phosphonic acid group in the HEDP was located at 530.9 eV (Figure S4b). In addition, the appearance of a peak (284.2 eV) in the C 1s spectrum of $Cu_2O@HEDP_{0.8}$ indicates the presence of a C−P bond (Figure S4c) [27]. These spectral results confirm the presence of the R−HPO_3^- structure on the surface of Cu_2O. Additionally, in the high-resolution Cu 2p spectrum of $Cu_2O@HEDP_{0.8}$, there were two peaks at 932.5 and 952.3 eV, corresponding to $2p_{3/2}$ and $2p_{1/2}$ of the Cu^+ species (Figure S4d), and with the Cu^{2+} peaks appeared at 934.3 and 954.2 eV. The proportion of Cu^{2+} increased slightly after HEDP modification, possibly due to the higher electronegativity of the additional O in HEDP [28]. This provides further evidence

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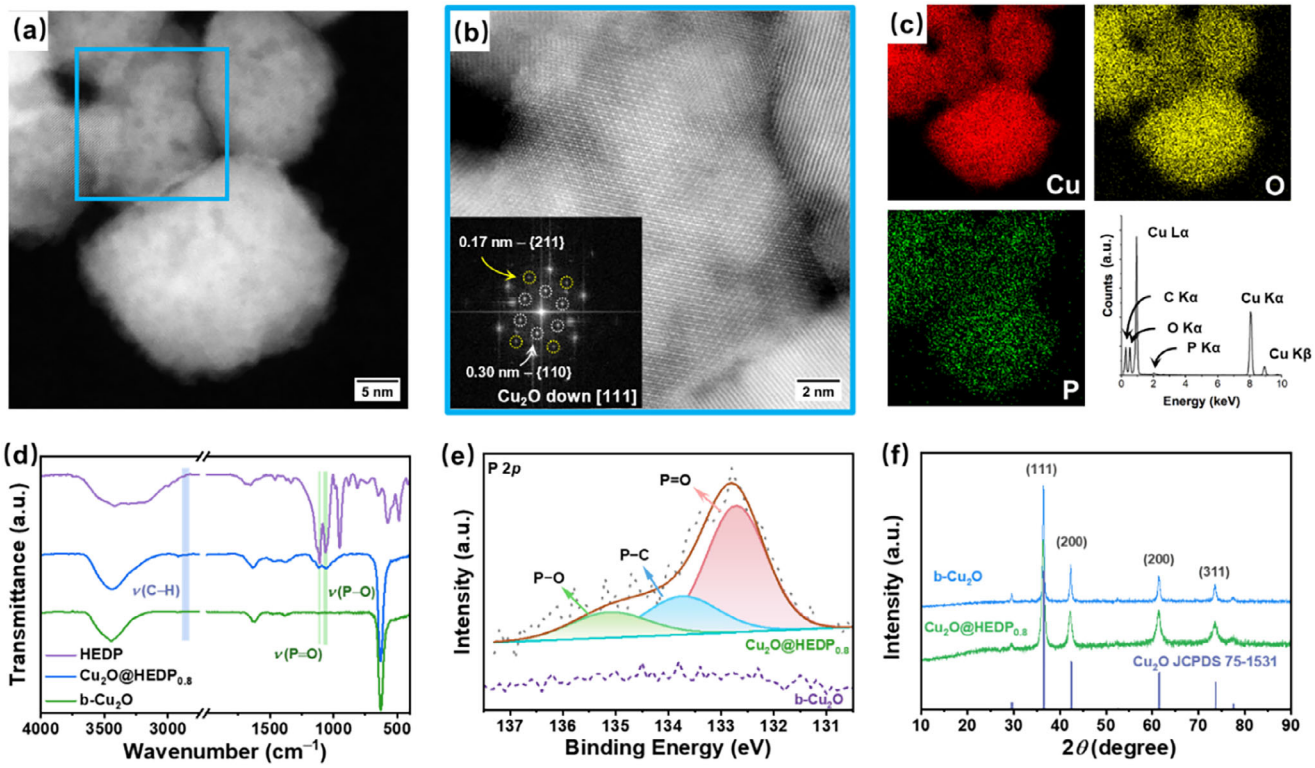

FIGURE 1 | Morphology and structural characterizations. (a) HAADF-STEM image of $Cu_2O@HEDP_{0.8}$ nanoparticles with crystalline regions. (b) Magnified view of the indicated area in (a) and a fast Fourier transform, indicating reflection symmetry suggesting Cu_2O viewed down the [111] axis. (c) EDS mapping images of $Cu_2O@HEDP_{0.8}$ and corresponding energy spectrum. (d) FT-IR spectra of HEDP, $Cu_2O@HEDP_{0.8}$, and b-Cu_2O. (e) High-resolution P 2p XPS spectra of $Cu_2O@HEDP_{0.8}$ and b-Cu_2O. (f) XRD patterns of $Cu_2O@HEDP_{0.8}$ and b-Cu_2O.

of an intrinsic interaction between the HEDP and Cu species. Notably, when the ion-beam etching depth reached ~40 nm, the P 2p signal was still observed (Figure S5), and the atomic content of P remained almost unchanged (Table S1), indicating that HEDP is not only distributed on the surface but is also present throughout the bulk phase of the catalyst.

The effect of the HEDP modification on the crystal structure was also investigated. Figure 1f shows the X-ray diffraction (XRD) patterns of $Cu_2O@HEDP_{0.8}$ and b-Cu_2O. The diffraction peaks at 36.5°, 42.4°, 61.5°, and 73.7° correspond to the (111), (200), (220), and (311) crystal facets of Cu_2O (PDF #05-0667), respectively [29]. The modification of HEDP does not affect the crystal structure of Cu_2O. The hydrophobicity of the catalysts was examined via water contact angle measurements (Figure S6). The angles for $Cu_2O@HEDP_{0.8}$ and b-Cu_2O were 21.5° and 21.0°, respectively. This small difference indicates that the introduction of HEDP did not have a significant impact on the surface hydrophobicity.

2.2 | Electrochemical Reduction Reaction

To verify the enhanced role of HEDP modification on the formation of C_{2+} products, we first evaluated the ECR performance of the $Cu_2O@HEDP_{0.8}$ and b-Cu_2O catalysts in an electrochemical flow cell, which consisted of a gas diffusion electrode (GDE), counter electrode (Pt sheet), and reference electrode (Ag/AgCl), with a 3.0 M KCl aqueous solution as the electrolyte (pH 1.0, Figure S7). The gaseous and liquid products were analyzed using gas chromatography (GC) and ^{1}H nuclear magnetic resonance (^{1}H-NMR) spectroscopy, respectively. The primary liquid products were C_3H_7OH, C_2H_5OH, CH_3COOH, and HCOOH, whereas the gaseous products were C_2H_4, CO, and H_2 (Figures S8 and S9).

The ECR performances of the $Cu_2O@HEDP_{0.8}$ and b-Cu_2O catalysts were investigated at various constant current densities ranging from −300 to −1100 mA cm^{-2}. Figure 2a illustrates the total current densities at various potentials over $Cu_2O@HEDP_{0.8}$ and b-Cu_2O. $Cu_2O@HEDP_{0.8}$ achieved the same current density at a lower potential than b-Cu_2O, indicating its superior catalytic activity. For instance, $Cu_2O@HEDP_{0.8}$ achieved a current density of −700 mA cm^{-2} at −1.94 V (vs RHE), whereas b-Cu_2O required −2.11 V (vs RHE) to reach the same current density. For both $Cu_2O@HEDP_{0.8}$ and b-Cu_2O, the primary C_{2+} products were C_2H_4 and C_2H_5OH, along with minor amounts of CH_3COOH and C_3H_7OH (Figure 2b,c). The ECR product distribution of the $Cu_2O@HEDP_{0.8}$ catalyst showed a continuous increase in FE_{C2+} within the current density range of −300 to −700 mA cm^{-2}. At −700 mA cm^{-2}, the FE_{C2+} over $Cu_2O@HEDP_{0.8}$ reached its peak value of 79.3% (Figure 2d), with C_2H_4 and C_2H_5OH FEs of 34.4% and 38.2%, respectively. However, FE_{H2} was significantly suppressed to below 5%. In contrast, FE_{C2+} catalyzed by b-Cu_2O at the same current density was only 47.0%, and FE_{H2} was up to ~33%. Notably, after the HEDP modification, the FE of C_2H_5OH increased significantly from 13.9% to 38.2% at a current density of −700 mA cm^{-2} (Figure S10). The partial current density of the C_{2+} products (j_{C2+}) over $Cu_2O@HEDP_{0.8}$ was significantly higher than that of b-Cu_2O (Figure 2e). At −900 mA cm^{-2}, the j_{C2+} over

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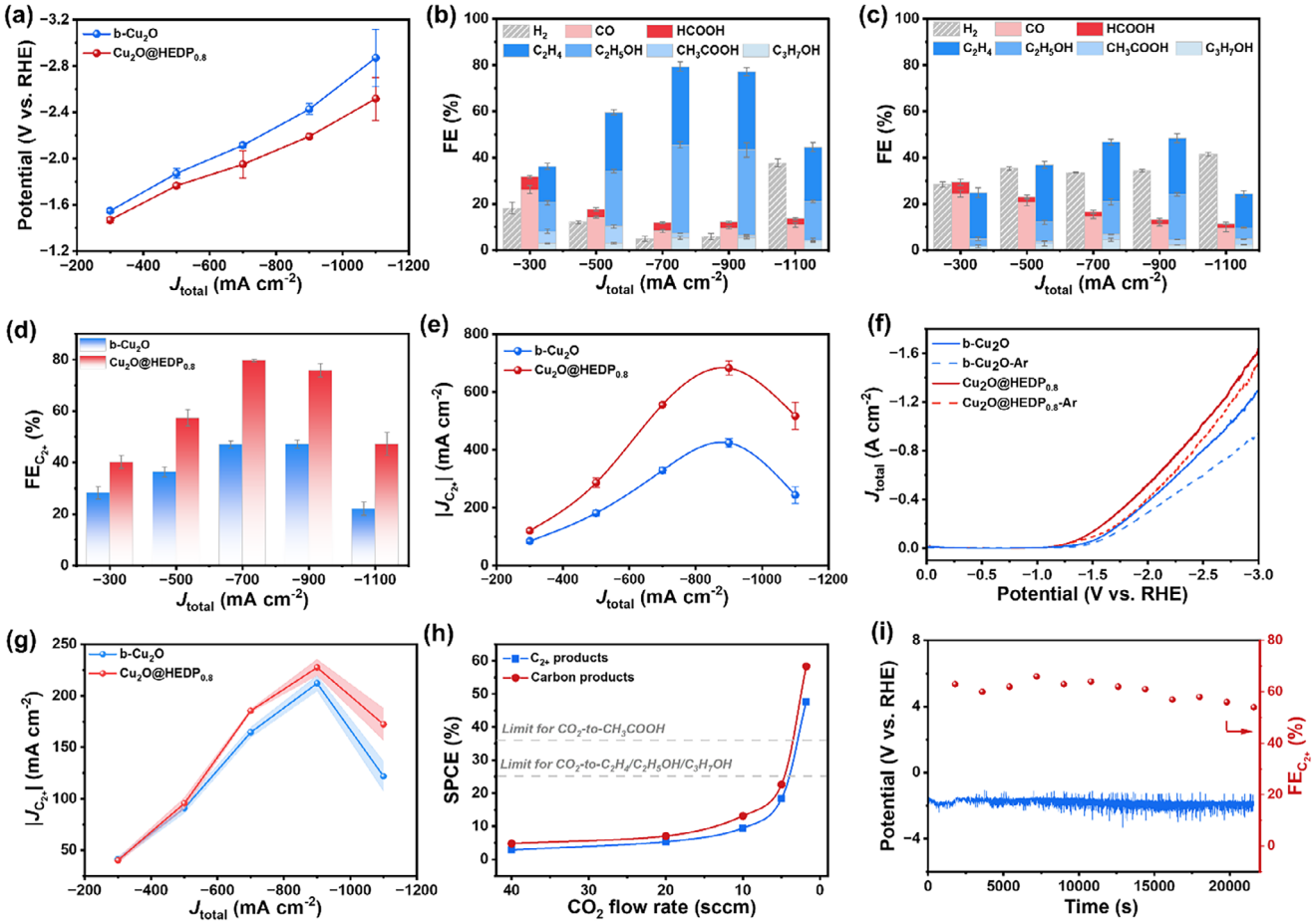

FIGURE 2 | ECR measurements over $Cu_2O@HEDP_{0.8}$ and b-Cu_2O in a flow cell. (a) Total current densities at various potential after iR compensation. (b) Product distribution under different applied current densities of $Cu_2O@HEDP_{0.8}$. (c) Product distribution under different applied current densities of b-Cu_2O. (d) Comparison of the FE_{C2+} under different applied current densities. (e) Comparison of the partial current density for C_{2+} products formation. (f) LSV curves in CO_2 and Ar atmosphere. (g) ECSA-normalized the j_{C2+} values. (h) SPCE of carbon products and C_{2+} products on $Cu_2O@HEDP_{0.8}$ under different CO_2 flow rates. (i) Stability test of the $Cu_2O@HEDP_{0.8}$ catalyst at a constant applied current density of −500 mA cm^{-2}. The error bars in the figure represent three repeated experiments.

$Cu_2O@HEDP_{0.8}$ reached −682.3 mA cm^{-2}, whereas that j_{C2+} for b-Cu_2O was only −423.2 mA cm^{-2}. Therefore, the $Cu_2O@HEDP_{0.8}$ catalyst exhibited significantly enhanced FE_{C2+} and j_{C2+} values at high current densities. Besides, the CO_2-to-C_{2+} performance of the $Cu_2O@HEDP_{0.8}$ catalyst surpasses that of recently reported catalysts in an acidic environment (Figure S11) [18, 30–35]. A detailed comparison with experimental conditions is shown in Table S2.

To investigate the impact of HEDP loading on ECR performance, we synthesized two additional $Cu_2O@HEDP$ catalysts with different amounts of HEDP (0.4 and 1.2 g), designated as $Cu_2O@HEDP_{0.4}$ and $Cu_2O@HEDP_{1.2}$, respectively. Both samples were shown to possess similar crystal structures and morphologies to the above $Cu_2O@HEDP_{0.8}$ (Figures S12 and S13). The product distributions of $Cu_2O@HEDP_{0.4}$ and $Cu_2O@HEDP_{1.2}$ at various current densities are displayed in Figure S14a,b, respectively. The FE_{C2+} at −700 mA cm^{-2} is seen to follow a volcano-shaped trend as a function of HEDP loading (Figure S14c). This result indicates that an optimal HEDP coverage effectively modulates the interfacial microenvironment to favor C_{2+} formation, rather than acting as an independent catalytic phase. Conversely, excessive HEDP loading leads to dense surface coverage that partially masks Cu active sites, resulting in a decline in FE_{C2+}.

To understand the enhanced ECR performance of the $Cu_2O@HEDP_{0.8}$ catalyst, linear sweep voltammetry (LSV) curves were obtained (Figure 2f). $Cu_2O@HEDP_{0.8}$ achieved a higher current density at the same potential, suggesting that $Cu_2O@HEDP_{0.8}$ possesses higher catalytic activity. Furthermore, $Cu_2O@HEDP_{0.8}$ exhibits a lower onset potential (−1.34 V vs RHE) under CO_2 flow than under Ar flow (−1.52 V vs RHE), indicating that $Cu_2O@HEDP_{0.8}$ was more inclined to undergo CO_2 reduction at low overpotentials (Figure S15). This demonstrates that the incorporation of HEDP into Cu_2O significantly improved its catalytic performance for CO_2 reduction. We further measured the electrochemical surface area (ECSA) of the $Cu_2O@HEDP_{0.8}$ and b-Cu_2O catalysts (Figure S16) and normalized the j_{C2+} values accordingly (Figure 2g). Even after normalization, the j_{C2+} of $Cu_2O@HEDP_{0.8}$ remained higher than that of b-Cu_2O, indicating that the enhanced ECR performance of $Cu_2O@HEDP_{0.8}$ originated from its intrinsic activity rather than the ECSA. As determined by electrochemical impedance

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spectroscopy (EIS), the interfacial charge-transfer resistances (R_{ct}) of $Cu_2O@HEDP_{0.8}$ and b-Cu_2O are 0.48 and 0.52 Ω, respectively (Figure S17). These closely matched values indicate that the introduction of HEDP does not cause a significant impact on electron transfer at the interface. Performing ECR in acidic electrolytes can mitigate the formation of (bi-) carbonates, thereby avoiding the CO_2 utilization limitations commonly observed in alkaline and neutral media [16, 36]. By reducing the CO_2 flow rate from 40 standard cubic centimeters per minute (sccm) to 1.8 sccm (Figure S18), the SPCE of carbon products increased from 4.7% to 58.4%. In addition, the SPCE of the C_{2+} products increased from 3.2% to 47.7%, breaking the limitation of the SPCE below 25% for C_{2+} (the limitation of CH_3COOH is 36%) generation in alkaline electrolytes (Figure 2h) [11].

The stability of $Cu_2O@HEDP_{0.8}$ was evaluated at a current density of −500 mA cm^{-2}. As shown in Figure 2i, C_{2+} FE exhibited no significant decline during the 6-h stability test. The morphology of $Cu_2O@HEDP_{0.8}$ after a 6-h ECR test was analyzed by STEM. No obvious morphological changes were observed for $Cu_2O@HEDP_{0.8}$ (Figure S19). EDS mapping showed that P element retains a uniform distribution across the catalyst after the stability test (Figure S20). Attenuated total reflectance Fourier transform infrared spectroscopy (ATR-IR) revealed that the phosphate groups remained on the catalyst surface after the ECR tests (Figure S21). The XRD patterns of $Cu_2O@HEDP_{0.8}$ after the ECR suggest that Cu_2O was converted into metallic Cu during the reaction, with a small amount of residual Cu_2O diffraction peaks attributed to surface oxidation (Figure S22). XPS was used to further analyze the surface chemical states of the catalysts. The Cu Auger spectrum also suggested that Cu_2O was converted into metallic Cu after the ECR test (Figure S23). Additionally, the R−HPO_3^- structure remained intact after the ECR test (Figure S24), which is consistent with the ATR-IR results. This suggests that HEDP can stably exist on the catalyst surface during the ECR reaction. Based on these results, we propose that the excellent ECR activity and stability of the $Cu_2O@HEDP_{0.8}$ catalyst can be attributed to the modified HEDP molecules. The presence of HEDP facilitated the enrichment of K^+ in the electric double layer and altered the interfacial H_2O molecular network structure, thereby promoting the ECR reaction and simultaneously suppressing the HER under acidic conditions.

2.3 | Mechanism Investigation

To verify the ability of HEDP to enhance interfacial K^+, we tested the ECR performance of $Cu_2O@HEDP_{0.8}$ and b-Cu_2O in electrolytes with K^+ concentrations of 0.05 M, 0.1 M, 1.0 M, and 3.0 M, respectively (Figure 3a; Figure S25). As the K^+ concentration in the electrolyte decreased, the FE of C_{2+} products over $Cu_2O@HEDP_{0.8}$ dropped by <15%. When the K^+ concentration in the electrolyte was only 0.05 M, the FE_{C2+} remained at 63.9%, indicating that the HEDP-modified catalyst can achieve efficient conversion of C_{2+} products in an electrolyte with low K^+ concentration. In contrast, the C_{2+} FE over b-Cu_2O significantly decreases, with an FE_{C2+} of only approximately 13% under 0.05 M K^+ conditions. To further confirm that the introduction of HEDP effectively enriches K^+ on the Cu_2O surface, we activated $Cu_2O@HEDP_{0.8}$ and b-Cu_2O in a 3.0 M K^+ electrolyte, then transferred the activated catalysts to 0.05 M H_2SO_4 for ECR performance evaluation. The activated $Cu_2O@HEDP_{0.8}$ still achieved 18% FE_{C2+} in the absence of K^+ in the electrolyte, whereas the b-Cu_2O catalyst did not produce C_{2+} products. This phenomenon supports our hypothesis that HEDP modification may enhance local K^+ concentrations, thereby promoting the generation of C_{2+} products.

Next, we used inductively coupled plasma optical emission spectroscopy (ICP-OES) to evaluate the local K^+ at the interfaces of $Cu_2O@HEDP_{0.8}$ and b-Cu_2O (Figure 3b). To avoid possible interference from Nafion-derived sulfonate groups, PTFE was employed as the binder during these measurements. The ICP-OES measurements yield a reproducible K^+ signal of 0.43 mg L^{-1} for $Cu_2O@HEDP_{0.8}$ under the defined polarized-transfer protocol, which is higher than that obtained for b-Cu_2O (Figure 3c). After normalization by the electrochemically active surface area, $Cu_2O@HEDP_{0.8}$ still exhibits a higher comparative interfacial K^+ association signal than b-Cu_2O (Figure S26), indicating that differences in surface area alone cannot account for the observed disparity. While a direct stoichiometric correlation between K^+ accumulation and HEDP loading would offer valuable mechanistic insight, reliable quantification of surface-bound HEDP is not currently feasible. Accordingly, the analysis focuses on comparative trends.

To investigate the mechanism behind the enhanced C_{2+} product formation performance of $Cu_2O@HEDP_{0.8}$ in acidic electrolytes, the mass transport kinetics of K^+ and H^+ within the reaction microenvironment were studied using a rotating disk electrode (RDE) in 0.05 M K_2SO_4 (pH 1.0) solution. $Cu_2O@HEDP_{0.8}$ exhibited a lower current density under an Ar atmosphere, indicating significant suppression of the HER. Additionally, both $Cu_2O@HEDP_{0.8}$ (Figure 3d) and b-Cu_2O (Figure 3e) showed a current plateau in the potential range from −0.9 to −1.5 V (vs RHE), suggesting that HER in this range is predominantly limited by H^+ diffusion [31, 37]. To verify the correlation between the plateau current and diffusion limitation, the RDE rotation rate was increased to enhance H^+ mass transport. Koutecky–Levich analysis shows that after modification with HEDP, the HER kinetic current density (j_k) decreases from 129.2 to 19.4 mA cm^{-2} (Figure 3f; Figure S27), indicating suppressed HER kinetics upon HEDP modification. In situ Raman spectroscopy was further employed to investigate the local pH (Figure S28), and the results indicate that the $Cu_2O@HEDP_{0.8}$ catalyst exhibited a higher local pH during the ECR process (Figures S29 and S30). These results above confirm our previous hypothesis that HEDP molecules may induce a high K^+ microenvironment. The accumulation of hydrated K^+ in electric double layer acts as an electrostatic shield, restricting the transport of H^+ from the bulk solution to the electrode surface, resulting in a lower current plateau [38].

In addition, in situ attenuated total reflection surface-enhanced infrared absorption spectroscopy (ATR-SEIRAS) was employed to investigate the structure of interfacial water molecules on the catalyst surface. The ν(O-H) stretching vibration bands representing the interfacial water structure (3800–3000 cm^{-1}) [39] were analyzed in detail. According to previous studies [40, 41], the O−H stretching pattern of H_2O can be deconvoluted into three peaks corresponding to four-coordinated hydrogen-bonded water (4HB-H_2O) at ~3250 cm^{-1}, weakly hydrogen-bonded water

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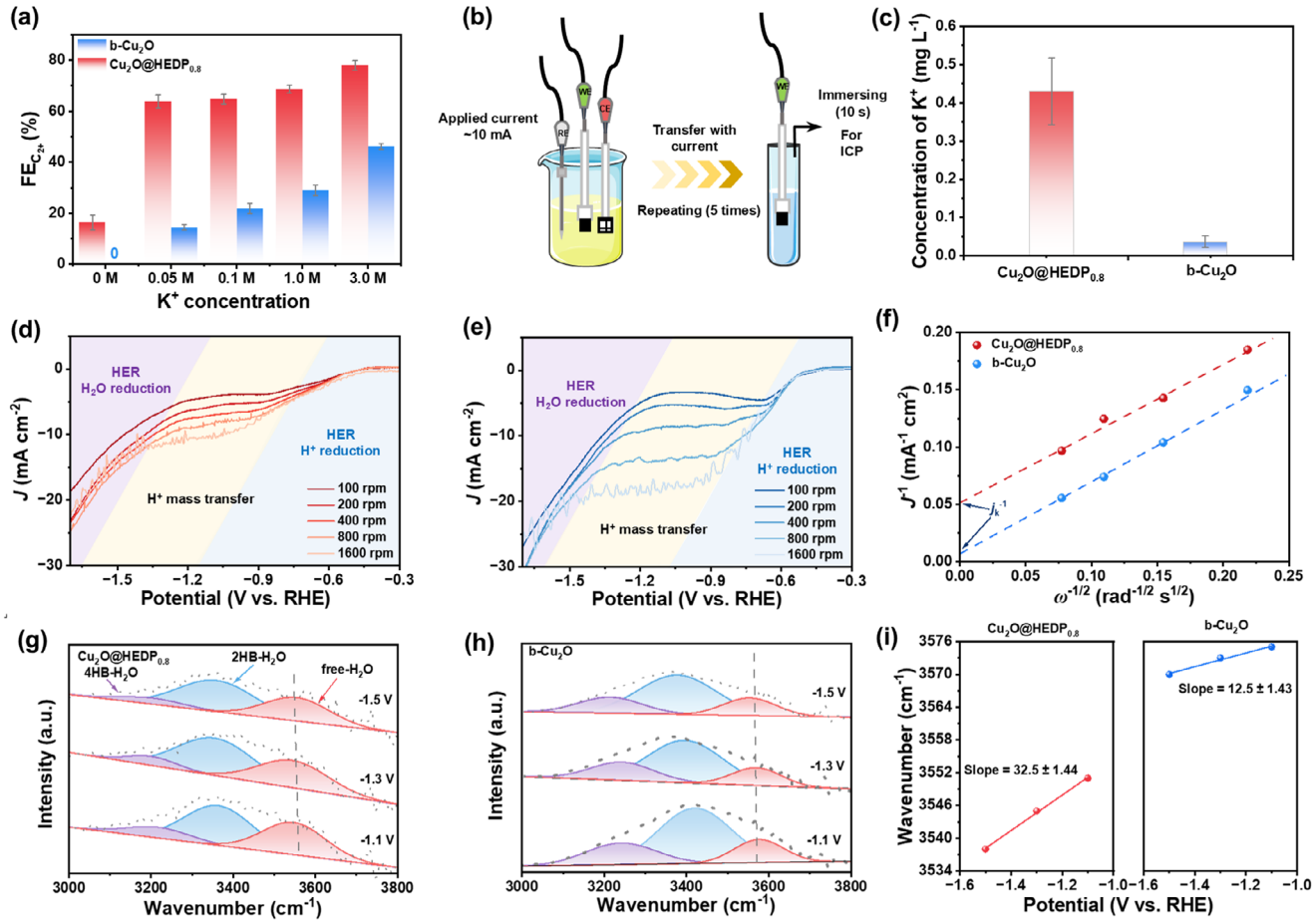

FIGURE 3 | Effect of K^+. (a) Comparison of FE_{C2+} at −700 mA cm^{-2} in pH 1.0 electrolytes with different K^+ concentrations. (b) Schematic diagram of the local K^+ concentration test on the catalyst surface. (c) Comparison of surface-associated/interfacial K^+ concentration on $Cu_2O@HEDP_{0.8}$ and b-Cu_2O. The error bars represent three independent experiments. LSV curves of (d) $Cu_2O@HEDP_{0.8}$ (e) b-Cu_2O in Ar-saturated 0.05 M K_2SO_4 (pH 1.0) under varied rotating speed. (f) Determination of the kinetic current density for HER through linear fitting of j^{-1} versus $\omega^{-1/2}$ for b-Cu_2O and $Cu_2O@HEDP_{0.8}$ based on Koutecky–Levich equation. ν(O-H) bands for the (g) $Cu_2O@HEDP_{0.8}$ and (h) b-Cu_2O at different potentials (vs RHE). (i) Stark tuning behavior of b-Cu_2O and $Cu_2O@HEDP_{0.8}$.

with two hydrogen bonds (2HB-H_2O) at ~3400 cm^{-1}, and free water molecules with very low degrees of hydrogen bonding (free H_2O) at ~3600 cm^{-1} (Figure 3g,h). With a negative shift in the potential, the proportion of free H_2O species on the surface of the $Cu_2O@HEDP_{0.8}$ catalyst increased and was significantly higher than that of b-Cu_2O (Figure S31). To investigate the origin of this change in the fraction of free water, we performed ATR-SEIRAS measurements on b-Cu_2O at −1.0 V (vs RHE). Upon injecting 50 µL of 60% HEDP solution into the electrolyte, the intensity of the ν(O—H) band at ~3450 cm^{-1} increased concomitantly (Figure S32). This suggests that the introduction of HEDP primarily affects 2HB-H_2O rather than free water. Therefore, combined with the ICP-OES results, the variation in free-water content is consistent with an altered interfacial ion/water microenvironment, in which interfacial K^+ may contribute.

To further explore the difference in the interfacial K^+ enrichment mechanisms between $Cu_2O@HEDP_{0.8}$ and b-Cu_2O, the Stark tuning behavior of free H_2O was analyzed. The results showed that the Stark tuning slope of free H_2O in $Cu_2O@HEDP_{0.8}$ was 32.5 cm^{-1} V^{-1}, which was significantly greater than that in b-Cu_2O, indicating that the free H_2O at the $Cu_2O@HEDP_{0.8}$ interface was in a stronger local electric field (Figure 3i) [42]. This result indicates that the introduction of HEDP leads to changes in the interfacial ionic environment.

Moreover, in situ ATR-SEIRAS was also performed for the ECR process catalyzed by $Cu_2O@HEDP_{0.8}$ (Figure 4a) and b-Cu_2O (Figure 4b) from −0.3 V to −1.6 V (vs RHE). The peak at 2050 cm^{-1} in Figure 4a is attributed to the atop configuration of the *CO intermediate, revealing the activation and reduction of CO_2. The red-shift of the *CO peaks with increasing cathodic potential results from the Stark effect [43], suggesting a stronger interaction between *CO and the catalyst [44]. The less obvious characteristic peaks near 1574 and 1187 cm^{-1} can be classified as *OCCO and *OCCOH intermediates [45]. Figure 4c illustrates the trends in the intensity variations of *CO and *OCCO over $Cu_2O@HEDP_{0.8}$. At −1.3 V (vs RHE), the *CO adsorption peak intensity reaches its highest value, and as the potential becomes more negative, the *CO intensity decreases. For *OCCO, the intensity gradually increases as the potential becomes more negative. In contrast, on the b-Cu_2O surface, the intensity of the *CO adsorption peak reached its maximum at −0.5 V (vs RHE), while the intensity of the *OCCO adsorption peak showed little variation and remained

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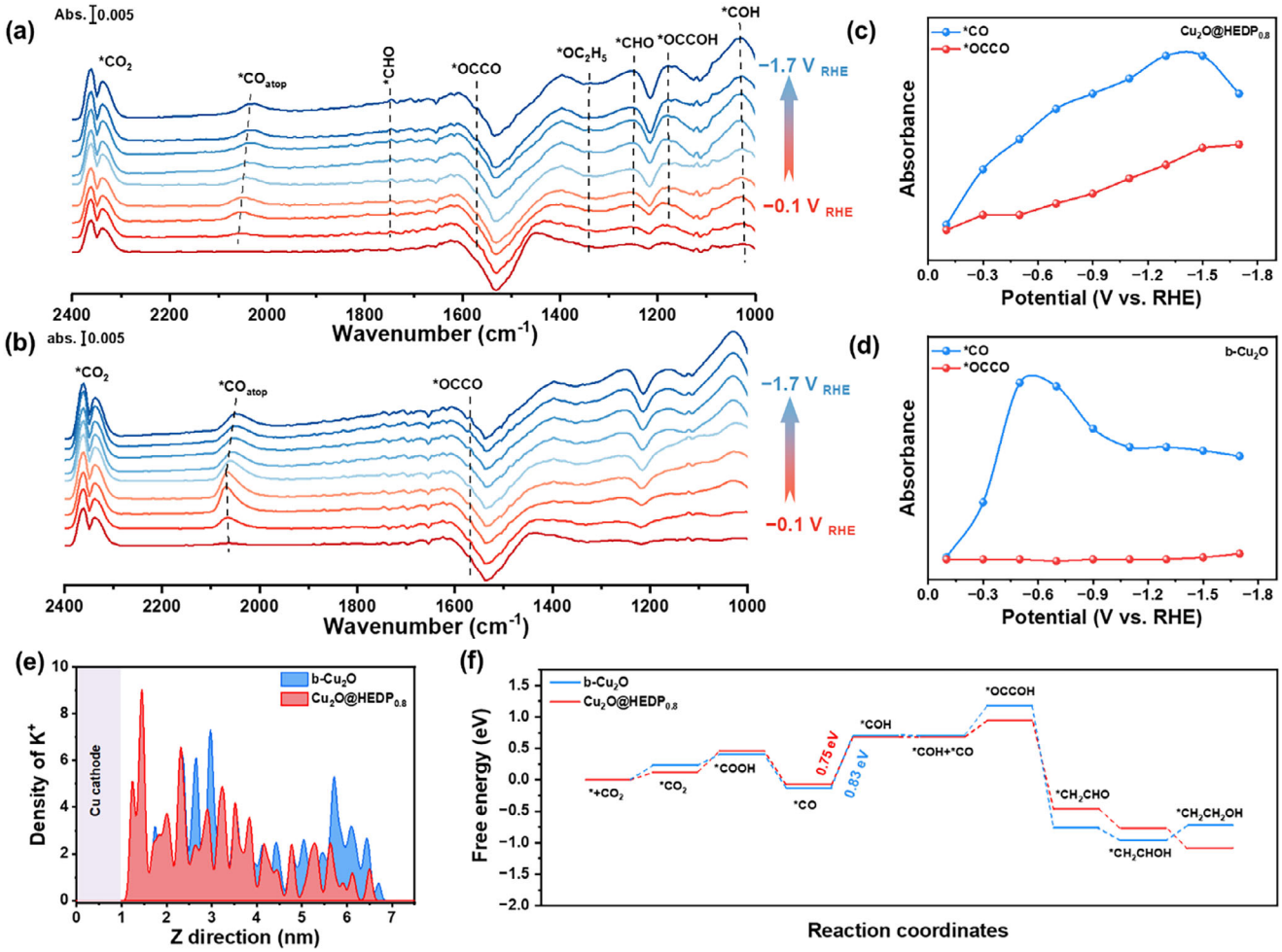

FIGURE 4 | Mechanism investigations. (a) In situ ATR-SEIRAS spectra of $Cu_2O@HEDP_{0.8}$ with changing cathodic potential. (b) In situ ATR-SEIRAS spectra of b-Cu_2O. (c) Absorbance of *CO and *OCCO at various potentials on $Cu_2O@HEDP_{0.8}$. (d) Absorbance of *CO and *OCCO at various potentials on b-Cu_2O. (e) Comparison of simulated density of K^+ versus distance from the Cu cathode. (f) Calculated Gibbs free energy diagram in CO_2 reduction reaction on $Cu_2O@HEDP_{0.8}$ and b-Cu_2O.

much lower than that of *CO (Figure 4d). This trend in the intensity variations of *CO and *OCCO may be due to the gradual accumulation of *CO on the $Cu_2O@HEDP_{0.8}$ surface as the potential became negative, which subsequently underwent C—C coupling to form *OCCO. On the surface of b-Cu_2O, most of the generated *CO intermediates desorbed from the surface. These observations are consistent with the previously observed ECR performance, where the FE_{CO} of b-Cu_2O was higher than that of $Cu_2O@HEDP_{0.8}$. Furthermore, key ethanol intermediates (*OC_2H_5) were detected at approximately 1346 cm^{-1}, confirming the presence of the ethanol pathway for $Cu_2O@HEDP_{0.8}$ [46].

Classical molecular dynamics (MD) simulations were employed to gain a deeper understanding at the molecular level of the influence of HEDP modification on the distribution of K^+ and CO_2 near the surface of the catalyst. We calculated the distribution of K^+ and CO_2 near the cathodes under a −0.1 V/Å electric field (Figure 4e; Figure S33). The densities of K^+ and CO_2 on the surface of $Cu_2O@HEDP_{0.8}$ were higher than those of b-Cu_2O. Consistent with the results observed by ATR-SEIRAS, the introduction of HEDP helped to promote the association of K^+ at the interface and bring more CO_2 molecules close to the cathode surface. The K^+-associating ability of HEDP enabled the $Cu_2O@HEDP_{0.8}$ catalyst to maintain excellent C_{2+} product generation in electrolytes with low K^+ concentrations.

Density functional theory (DFT) calculations were employed to gain a deeper understanding of the reaction mechanism for ECR to C_{2+} products over $Cu_2O@HEDP_{0.8}$ (Figure 4f). The corresponding computational models are presented in Figures S34–S36. The first step in the reaction pathway for generating C_{2+} products on copper surfaces is the activation of the linear CO_2 molecule into the adsorbed CO_2 (*CO_2) species, which is then converted into the key C_1 intermediate *CO via the *COOH intermediate [47, 48]. The Gibbs free energy changes (ΔG) for CO_2 activation on $Cu_2O@HEDP_{0.8}$ and b-Cu_2O were 0.12 and 0.23 eV, respectively. This indicated that modification with HEDP was more favorable for CO_2 activation.

The Gibbs free energy for the generation of C_{2+} products was calculated based on the *OCCO and *OCCOH intermediates observed using ATR-SEIRAS spectroscopy. In the free energy diagram, the conversion of *CO to *COH was found to be the rate-determining step (RDS). For $Cu_2O@HEDP_{0.8}$, the ΔG value

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for this conversion was 0.75 eV, which was lower than that of b-Cu_2O (0.83 eV). The decrease in the *CO protonation energy barrier may be attributed to the localized high K^+ environment induced by HEDP modification, which weakens the H−OH bond and promotes H_2O dissociation to form *H [49]. The subsequent C−C coupling step was found to be the key step in the formation of C_{2+} products. Over Cu_2O@$HEDP_{0.8}$ and b-Cu_2O, the energy barriers for the coupling of *CO and *COH to form *OCCOH were 0.20 and 0.41 eV, respectively. Furthermore, for Cu_2O@$HEDP_{0.8}$, the protonation of *OCCOH to form *CH_2CH_2OH occurs spontaneously, whereas for b-Cu_2O, an additional energy barrier must be overcome. This further indicates that the Cu_2O@HEDP surface is more favorable for C_2H_5OH formation, which aligns with the observed increase in FE_{C2H5OH} during ECR reactions. These DFT calculations suggest a qualitative trend in which HEDP modification lowers the free-energy barrier for key steps, providing complementary insight that is consistent with the experimental observations.

Based on our experimental and theoretical calculation results, we propose a possible mechanism for the enhanced formation of C_{2+} products induced by HEDP modification (Figure S37). First, the association between $-HPO_3^-$ and K^+ leads to local K^+ enrichment at the interface. The interaction between K^+ and the water molecules weakens the H−OH bond in H_2O, thereby promoting the hydrogenation of *CO to *COH. Subsequently, the C−C coupling step was facilitated by the high local pH induced by the high K^+ microenvironment. Thus, although the K^+ concentration in the electrolyte was relatively low in this study, the process of electrocatalytic CO_2 reduction to C_{2+} products can still be effectively promoted under acidic conditions because of the locally confined K^+ microenvironment at the interface.

2.4 | Demonstration in K^+-Free Electrolyte

Based on the great potential of Cu_2O@$HEDP_{0.8}$ in low K^+ acidic ECR, its feasibility for efficient CO_2 electroconversion in acidic electrolytes without K^+ was evaluated. A membrane electrode assembly (MEA) electrolyzer with an active area of 1 cm × 1 cm was assembled using the prepared Cu_2O@$HEDP_{0.8}$ electrode as the cathode and IrO_2-loaded Ti mesh as the anode (Figure 5a). Prior to ECR performance testing, the anode chamber was first filled with 0.5 M K_2SO_4 electrolyte (pH 4.0) for catalyst activation at an applied voltage of 4.0 V. Subsequently, 1 mM H_2SO_4 was used as the anolyte, and the ECR performance was evaluated under applied voltages ranging from 4.0 to 4.4 V (Figure 5b). At a cell voltage of 4.2 V, the FE for C_{2+} products reached approximately 50%, whereas the FE for H_2 was maintained at approximately 15% (Figure 5c). Under a voltage of 4.1 V, long-term electrolysis tests demonstrated that the Cu_2O@$HEDP_{0.8}$ catalyst could operate stably for 20 h, with only slight fluctuations in FE_{C2+} (Figure 5d). The local and selective enrichment of K^+ on the Cu_2O@$HEDP_{0.8}$ catalyst significantly reduced the operating costs and improved the catalytic stability. Although operation under acidic conditions with low K^+ concentration requires higher cell voltages, this suppresses carbon loss. Additionally, rational interfacial regulation can partially mitigate these challenges within the limits of thermodynamic and mass-transport constraints.

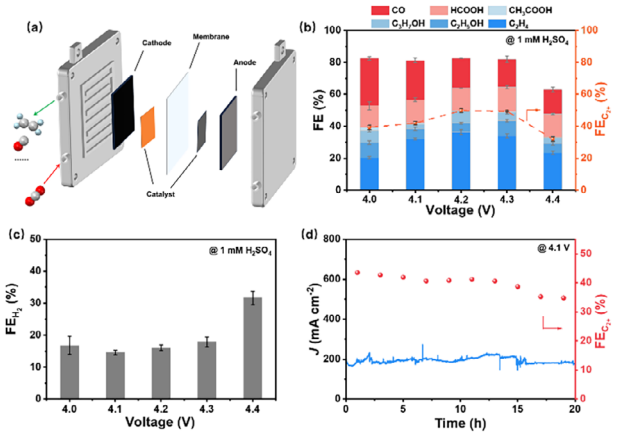

FIGURE 5 | (a) Schematic diagram of MEA. (b) Product distribution under different cell voltages of Cu_2O@$HEDP_{0.8}$. (c) FE_{H2} of Cu_2O@$HEDP_{0.8}$. (d) Stability test of Cu_2O@$HEDP_{0.8}$. The error bars in the figure represent two repeated experiments.

3 | Conclusions

This study demonstrates a strategy for modifying Cu_2O with HEDP to increase the K^+ concentration in the electrical double layer, enabling efficient CO_2 conversion to C_{2+} products under acidic conditions at K^+-free concentrations. The HEDP-modified Cu_2O catalyst achieved a maximum FE of 79.3% for C_{2+} products at a current density of −700 mA cm^{-2} in a 3.0 M K^+ electrolyte (pH 1.0). Remarkably, even in a K^+-free electrolyte (1 mM H_2SO_4), the FE for the C_{2+} products remained as high as 50%. In situ characterization and MD simulations confirmed the enrichment of K^+ in the double electric layer after HEDP modification. Further DFT calculations provide qualitative insight suggesting that HEDP may promote the activation of CO_2, facilitates the protonation of *CO intermediates, and enhances subsequent C−C coupling to generate C_{2+} products. This work presents a novel approach to enhancing the CO_2-to-C_{2+} conversion in acidic environments by introducing phosphonates to regulate the interfacial microenvironment. By tuning the distribution of local cations and the structure of interfacial water, the issue of salt precipitation can be effectively alleviated while enhancing C_{2+} selectivity. This strategy emphasizes the crucial role of the electrode-electrolyte interface, where rational catalyst design can effectively balance the K^+ concentration trade-offs in acidic ECR systems.

4 | Experimental Section

4.1 | Synthesis of b-Cu_2O

The Cu_2O catalyst was synthesized using a chemical reduction method. At first, $Na_2B_4O_7$ (2.0 g) was dissolved in hot deionized (70 mL) water at 90°C. Subsequently, as the $Na_2B_4O_7$ aqueous solution was cooled to room temperature, L-ascorbic acid (0.8 g) was added to the solution under vigorous stirring. After 15 min of vigorous stirring, $CuCl_2$ (0.4 g) was added to this solution. The product was collected by centrifugation, washed six times with deionized water, and dried under vacuum at 60°C for 12 h.

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4.2 | Synthesis of $Cu_2O@HEDP_{0.8}$

The $Cu_2O@HEDP_{0.8}$ catalyst was synthesized using a chemical reduction method. First, $Na_2B_4O_7$ (2.0 g) was dissolved in hot deionized (70 mL) water at 90°C. When the temperature of $Na_2B_4O_7$ aqueous solution was dropped to room temperature, L-ascorbic acid (0.8 g) and HEDP (0.8 g) were added into the $Na_2B_4O_7$ aqueous solution under vigorous stirring. After 15 min of vigorous stirring, $CuCl_2$ (0.4 g) was added to this solution. The product was collected by centrifugation, washed six times with deionized water, and dried under vacuum at 60°C for 12 h. $Cu_2O@HEDP_{0.4}$ and $Cu_2O@HEDP_{1.2}$ catalysts were synthesized using 0.4 and 1.2 g of HEDP, respectively.

Characterizations and Computational details are provided in Supporting Information.

Acknowledgements

This work was supported by the Joint Funds of the National Natural Science Foundation of China (U24B20201), the National Natural Science Foundation of China (22372007 and 21972010), and the Fundamental Research Funds for the Central Universities (JD2427). J.G.M. thanks the funding received from the Spanish Ministry of Science, Grant PID2021-1287610B-C21 by MCIN/AEI/10.13039/501100011033 through "ERDF A way of making Europe," the Generalitat Valenciana through CIPROM/2023/2, and the Advanced Materials Program and was supported by MCIN with funding from the European Union NextGenerationEU (PRTR-C17. I1) and Generalitat Valenciana. M.Y. acknowledges funding support from the Hong Kong Polytechnic University, Hong Kong SAR, China (project numbers: P0042711 and P0048122) and Guangdong Natural Science Foundation, China (project number: 2024A1515010031).

Funding

The Joint Funds of the National Natural Science Foundation of China (U24B20201), the National Natural Science Foundation of China (22372007 and 21972010), and the Fundamental Research Funds for the Central Universities (JD2427). Funding support from the Hong Kong Polytechnic University, China (P0042711 and P0048122). Guangdong Natural Science Foundation, China (2024A1515010031).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. X. Chen, J. Chen, N. M. Alghoraibi, et al., "Electrochemical CO_2-to-Ethylene Conversion on Polyamine-Incorporated Cu Electrodes," *Nature Catalysis* 4, no. 1 (2021): 20–27, https://doi.org/10.1038/s41929-020-00547-0.

2. B. Seger, M. Robert, and F. Jiao, "Best Practices for Electrochemical Reduction of Carbon Dioxide," *Nature Sustainability* 6, no. 3 (2023): 236–238, https://doi.org/10.1038/s41893-022-01034-z.

3. A. Ozden, F. P. García de Arquer, J. E. Huang, et al., "Carbon-Efficient Carbon Dioxide Electrolysers," *Nature Sustainability* 5, no. 7 (2022): 563–573, https://doi.org/10.1038/s41893-022-00879-8.

4. X. Wang, P. Li, J. Tam, et al., "Efficient CO and Acrolein Co-Production via Paired Electrolysis," *Nature Sustainability* 7, no. 7 (2024): 931–937, https://doi.org/10.1038/s41893-024-01363-1.

5. Y. Yang, C. Zhang, C. Zhang, et al., "Ligand-tuning Copper in Coordination Polymers for Efficient Electrochemical C—C Coupling," *Nature Communications* 15, no. 1 (2024): 6316, https://doi.org/10.1038/s41467-024-50791-2.

6. X. Li, W. Kang, X. Fan, et al., "Electrochemical CO_2 Reduction to Liquid Fuels: Mechanistic Pathways and Surface/Interface Engineering of Catalysts and Electrolytes," *The Innovation* 6, no. 3 (2025): 100807, https://doi.org/10.1016/j.xinn.2025.100807.

7. Z. Yao, H. Cheng, Y. Xu, et al., "Hydrogen Radical-Boosted Electrocatalytic CO_2 Reduction Using Ni-Partnered Heteroatomic Pairs," *Nature Communications* 15, no. 1 (2024): 9881, https://doi.org/10.1038/s41467-024-53529-2.

8. M. Zhuansun, Y. Liu, R. Lu, et al., "Promoting CO_2 Electroreduction to Multi-Carbon Products by Hydrophobicity-Induced Electro-Kinetic Retardation," *Angewandte Chemie International Edition* 62, no. 41 (2023): 202309875, https://doi.org/10.1002/anie.202309875.

9. C. Chen, X. Yan, S. Liu, et al., "Highly Efficient Electroreduction of CO_2 to C_{2+} Alcohols on Heterogeneous Dual Active Sites," *Angewandte Chemie International Edition* 59, no. 38 (2020): 16459–16464, https://doi.org/10.1002/anie.202006847.

10. W. Liu, P. Zhai, A. Li, et al., "Electrochemical CO_2 Reduction to Ethylene by Ultrathin CuO Nanoplate Arrays," *Nature Communications* 13, no. 1 (2022): 1877, https://doi.org/10.1038/s41467-022-29428-9.

11. H. Deng, T. Liu, W. Zhao, et al., "Substituent Tuning of Cu Coordination Polymers Enables Carbon-Efficient CO_2 Electroreduction to Multi-Carbon Products," *Nature Communications* 15, no. 1 (2024): 9706, https://doi.org/10.1038/s41467-024-54107-2.

12. C. P. O'Brien, R. K. Miao, S. Liu, et al., "Single Pass CO_2 Conversion Exceeding 85% in the Electrosynthesis of Multicarbon Products via Local CO_2 Regeneration," *ACS Energy Letters* 6, no. 8 (2021): 2952–2959, https://doi.org/10.1021/acsenergylett.1c01122.

13. J. A. Rabinowitz and M. W. Kanan, "The Future of Low-Temperature Carbon Dioxide Electrolysis Depends on Solving One Basic Problem," *Nature Communications* 11, no. 1 (2020): 5231, https://doi.org/10.1038/s41467-020-19135-8.

14. J. Gu, S. Liu, W. Ni, W. Ren, S. Haussener, and X. Hu, "Modulating Electric Field Distribution by Alkali Cations for CO_2 Electroreduction in Strongly Acidic Medium," *Nature Catalysis* 5, no. 4 (2022): 268–276, https://doi.org/10.1038/s41929-022-00761-y.

15. L. Li, Z. Liu, X. Yu, and M. Zhong, "Achieving High Single-Pass Carbon Conversion Efficiencies in Durable CO_2 Electroreduction in Strong Acids via Electrode Structure Engineering," *Angewandte Chemie International Edition* 62, no. 21 (2023): 202300226, https://doi.org/10.1002/anie.202300226.

16. J. E. Huang, F. Li, A. Ozden, et al., "CO_2 Electrolysis to Multicarbon Products in Strong Acid," *Science* 372, no. 6546 (2021): 1074–1078, https://doi.org/10.1126/science.abg6582.

17. Y. Zhao, L. Hao, A. Ozden, et al., "Conversion of CO_2 to Multicarbon Products in Strong Acid by Controlling the Catalyst Microenvironment," *Nature Synthesis* 2, no. 5 (2023): 403–412, https://doi.org/10.1038/s44160-022-00234-x.

18. Z. Ma, Z. Yang, W. Lai, et al., "CO_2 Electroreduction to Multicarbon Products in Strongly Acidic Electrolyte via Synergistically Modulating the Local Microenvironment," *Nature Communications* 13, no. 1 (2022): 7596, https://doi.org/10.1038/s41467-022-35415-x.

19. S. Garg, Q. Xu, A. B. Moss, et al., "How Alkali Cations Affect Salt Precipitation and CO_2 Electrolysis Performance in Membrane Electrode Assembly Electrolyzers," *Energy & Environmental Science* 16, no. 4 (2023): 1631–1643, https://doi.org/10.1039/D2EE03725D.

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20. K. Popov, H. Rönkkömäki, and L. Lajunen, "Critical Evaluation of Stability Constants of Phosphonic Acids (IUPAC Technical Report)," *Pure and Applied Chemistry* 73, no. 10 (2002): 1641–1677, https://doi.org/10.1351/pac200173101641.

21. Y.-T. Yang, X.-H. Wang, Z.-N. Zhang, et al., "Au-Modified Cu_2O Nanocubes for Light-Enhanced Nitrate Electroreduction to Ammonia," *Journal of Materials Chemistry A* 12, no. 43 (2024): 29969–29977, https://doi.org/10.1039/D4TA06439A.

22. M. Othmani, A. Aissa, C. G. Bac, F. Rachdi, and M. Debbabi, "Surface Modification of Calcium Hydroxyapatite by Grafting of Etidronic Acid," *Applied Surface Science* 274, no. 1 (2013): 151–157, https://doi.org/10.1016/j.apsusc.2013.03.002.

23. W. Shi, M. Xia, F. Wang, L. Dong, and S. Zhu, "Efficient Absorption Properties of Surface Grafted HEDP-HAP Composites for Pb^{2+} and Cu^{2+}: Experimental Study and Visualization Study of Interaction Based on Becke Surface Analysis and Independent Gradient Model," *Journal of Hazardous Materials* 401, no. 5 (2021): 123748, https://doi.org/10.1016/j.jhazmat.2020.123748.

24. L. Dong, S. Zhu, M. Xia, Y. Chu, F. Wang, and W. Lei, "Molecular Dynamics Simulations of the Binging Affinity of 1-Hydroxyethane-1, 1-Diphosphonic Acid (HEDP) with Nano-Hydroxyapatite and the Uptake of Cu^{2+} by HEDP-HAP Hybrid Systems," *Journal of Hazardous Materials* 383, no. 5 (2020): 121206, https://doi.org/10.1016/j.jhazmat.2019.121206.

25. J.-S. Li, Y. Wang, C.-H. Liu, et al., "Coupled Molybdenum Carbide and Reduced Graphene Oxide Electrocatalysts for Efficient Hydrogen Evolution," *Nature Communications* 7, no. 1 (2016): 11204, https://doi.org/10.1038/ncomms11204.

26. P. Ma, R. Bi, Q. Wang, L. Lu, X. Ma, and F. Chen, "In Situ Synthesis Copper Phosphate-Protein Hybrid Nanoflower on Nickel Foam for the Sensitive Detection of Glucose in Body Fluids," *Materials Research Bulletin* 170 (2024): 112583, https://doi.org/10.1016/j.materresbull.2023.112583.

27. M. Y. Bashouti, Y. Paska, S. R. Puniredd, T. Stelzner, S. Christiansen, and H. Haick, "Silicon Nanowires Terminated with Methyl Functionalities Exhibit Stronger Si–C Bonds than Equivalent 2D Surfaces," *Physical Chemistry Chemical Physics* 11, no. 20 (2009): 3845–3848, https://doi.org/10.1039/B820559K.

28. J. Zhou, B. He, P. Huang, et al., "Regulating Interfacial Hydrogen-Bonding Networks by Implanting Cu Sites with Perfluorooctane to Accelerate CO_2 Electroreduction to Ethanol," *Angewandte Chemie International Edition* 64, no. 6 (2025): 202418459, https://doi.org/10.1002/anie.202418459.

29. J. Cheng, L. Wu, and J. Luo, "Improving the Photovoltage of Cu_2O Photocathodes with Dual Buffer Layers," *Nature Communications* 14, no. 1 (2023): 7228, https://doi.org/10.1038/s41467-023-42799-x.

30. J. Feng, L. Wu, X. Song, et al., "CO_2 Electrolysis to Multi-Carbon Products in Strong Acid at Ampere-Current Levels on La-Cu Spheres with Channels," *Nature Communications* 15, no. 1 (2024): 4821, https://doi.org/10.1038/s41467-024-49308-8.

31. W. Nie, G. P. Heim, N. B. Watkins, T. Agapie, and J. C. Peters, "Organic Additive-Derived Films on Cu Electrodes Promote Electrochemical CO_2 Reduction to C_{2+} Products under Strongly Acidic Conditions," *Angewandte Chemie International Edition* 62, no. 12 (2023): 202216102, https://doi.org/10.1002/anie.202216102.

32. Y. Yang, J. Zhang, Z. Tan, et al., "Highly Selective Production of C_{2+} Oxygenates from CO_2 in Strongly Acidic Condition by Rough Ag-Cu Electrocatalyst," *Angewandte Chemie International Edition* 63, no. 47 (2024): 202408873, https://doi.org/10.1002/anie.202408873.

33. X.-R. Qin, J.-J. Li, L.-L. Wang, et al., "Electroreduction of Diluted CO_2 to Multicarbon Products with High Carbon Utilization at 800 mA cm^{-2} in Strongly Acidic media," *Nature Communications* 16, no. 1 (2025): 4447, https://doi.org/10.1038/s41467-025-59783-2.

34. Y. Qiao, S. Shen, C. Mao, et al., "Interfacial Oxygen Vacancy-Copper Pair Sites on Inverse CeO_2/Cu Catalyst Enable Efficient CO_2 Electroreduction to Ethanol in Acid," *Angewandte Chemie International Edition* 64, no. 13 (2025): 202424248, https://doi.org/10.1002/anie.202424248.

35. Y. Wang, W. Lai, H. Tao, et al., "CO_2 Electroreduction to Multicarbon Products over Cu_2O@Mesoporous SiO_2 Confined Catalyst: Relevance of the Shell Thickness," *Advanced Energy Materials* 15, no. 16 (2025): 2404606, https://doi.org/10.1002/aenm.202404606.

36. M. Zeng, W. Fang, Y. Cen, X. Zhang, Y. Hu, and B. Y. Xia, "Reaction Environment Regulation for Electrocatalytic CO_2 Reduction in Acids," *Angewandte Chemie International Edition* 63, no. 26 (2024): 202404574, https://doi.org/10.1002/anie.202404574.

37. M. Fan, J. E. Huang, R. K. Miao, et al., "Cationic-group-functionalized Electrocatalysts Enable Stable Acidic CO_2 Electrolysis," *Nature Catalysis* 6, no. 9 (2023): 763–772, https://doi.org/10.1038/s41929-023-01003-5.

38. Z. Wang, D. Liu, C. Xia, et al., "Tip Carbon Encapsulation Customizes Cationic Enrichment and Valence Stabilization for Low K^+ Acidic CO_2 Electroreduction," *Nature Communications* 16, no. 1 (2025): 1754, https://doi.org/10.1038/s41467-025-56977-6.

39. Q. Sun, N. J. Oliveira, S. Kwon, et al., "Understanding Hydrogen Electrocatalysis by Probing the Hydrogen-bond Network of Water at the Electrified Pt–solution Interface," *Nature Energy* 8, no. 8 (2023): 859–869, https://doi.org/10.1038/s41560-023-01302-y.

40. R. R. Rao, B. Huang, Y. Katayama, et al., "pH- and Cation-Dependent Water Oxidation on Rutile RuO_2(110)," *The Journal of Physical Chemistry C* 125, no. 15 (2021): 8195–8207, https://doi.org/10.1021/acs.jpcc.1c00413.

41. Y.-H. Wang, S. Zheng, W.-M. Yang, et al., "In Situ Raman Spectroscopy Reveals the Structure and Dissociation of Interfacial Water," *Nature* 600, no. 7887 (2021): 81–85, https://doi.org/10.1038/s41586-021-04068-z.

42. Q. Zhu, C. L. Rooney, H. Shema, et al., "The Solvation Environment of Molecularly Dispersed Cobalt Phthalocyanine Determines Methanol Selectivity during Electrocatalytic CO_2 Reduction," *Nature Catalysis* 7, no. 9 (2024): 987–999, https://doi.org/10.1038/s41929-024-01190-9.

43. J. Zhang, J. Ding, Y. Liu, et al., "Molecular Tuning for Electrochemical CO_2 Reduction," *Joule* 7, no. 8 (2023): 1700–1744, https://doi.org/10.1016/j.joule.2023.07.010.

44. R. Zhang, J. Zhang, S. Wang, et al., "Synthesis of n-Propanol from CO_2 Electroreduction on Bicontinuous Cu_2O/Cu Nanodomains," *Angewandte Chemie International Edition* 63, no. 30 (2024): 202405733, https://doi.org/10.1002/anie.202405733.

45. Y. Kim, S. Park, S.-J. Shin, et al., "Time-Resolved Observation of C–C Coupling Intermediates on Cu Electrodes for Selective Electrochemical CO_2 Reduction," *Energy & Environmental Science* 13, no. 11 (2020): 4301–4311, https://doi.org/10.1039/D0EE01690J.

46. F. Ma, P. Zhang, X. Zheng, et al., "Steering the Site Distance of Atomic Cu–Cu Pairs by First-Shell Halogen Coordination Boosts CO_2-to-C_2 Selectivity," *Angewandte Chemie International Edition* 63, no. 46 (2024): 202412785, https://doi.org/10.1002/anie.202412785.

47. T. K. Todorova, M. W. Schreiber, and M. Fontecave, "Mechanistic Understanding of CO_2 Reduction Reaction (CO_2RR) toward Multicarbon Products by Heterogeneous Copper-Based Catalysts," *ACS Catalysis* 10, no. 3 (2020): 1754–1768, https://doi.org/10.1021/acscatal.9b04746.

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48. X. He, L. Lin, X. Li, et al., "Roles of Copper(I) in Water-Promoted CO_2 Electrolysis to Multi-Carbon Compounds," *Nature Communications* 15, no. 1 (2024): 9923, https://doi.org/10.1038/s41467-024-54282-2.

49. Z. Wei, J. Ding, and Z. Wang, "Enhanced Electrochemical CO_2 Reduction to Formate over Phosphate-Modified In: Water Activation and Active Site Tuning," *Angewandte Chemie, International Edition* 63, (2024), 202402070, https://doi.org/10.1002/anie.202402070.

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